

Predicting open system quantum trajectories with deep learning

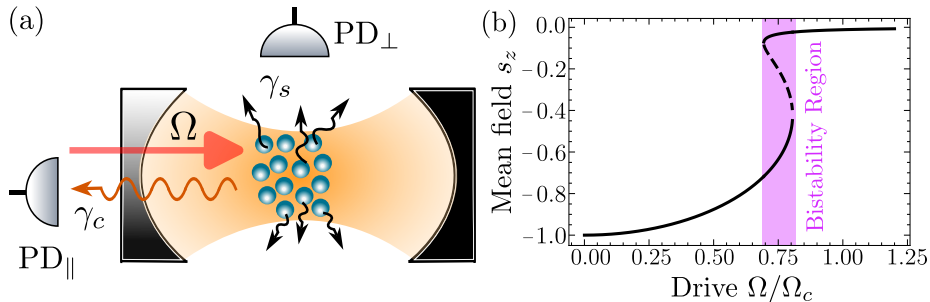
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- 1 Proposal
- 2 Fixed N and Ω : Generating more trajectories
- 3 Dataset

Physics (short): Driven-dissipative cavity QED



(a) System under study (b) Mean-field solution and bistability region

Main equation – Master equation

$$\dot{\rho} = -i[\mathcal{H}, \rho] + \frac{\gamma_c}{2} \mathcal{D}[\hat{S}_-]\rho + \frac{\gamma_s}{2} \sum_n \mathcal{D}[\hat{\sigma}_n]\rho, \quad \mathcal{H} = 2\Omega\hat{S}_x \quad (1)$$

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How to solve the master equation? Answer: MCWF method

Generally: Solve for density matrix: $4^N \times 4^N$ elements, N is the number of qubits.
Instead: Probabilistic unraveling of the master equation into quantum trajectories

$$|\psi(t + dt)\rangle = \begin{cases} \frac{\hat{S}_- |\psi(t)\rangle}{\sqrt{p_-}} & \text{with probability } p_- = \gamma_c dt \langle \psi(t) | \hat{S}_+ \hat{S}_- | \psi(t) \rangle \\ \frac{\hat{\sigma}_n |\psi(t)\rangle}{\sqrt{p_n}} & \text{with probability } p_n = \gamma_s dt \langle \psi(t) | \hat{\sigma}_n^\dagger \hat{\sigma}_n | \psi(t) \rangle \\ \frac{e^{-i\mathcal{H}_{NH}dt} |\psi(t)\rangle}{\sqrt{p_0}} & \text{with probability } p_0 = 1 - p_- - \sum_n p_n \end{cases} \quad (2)$$

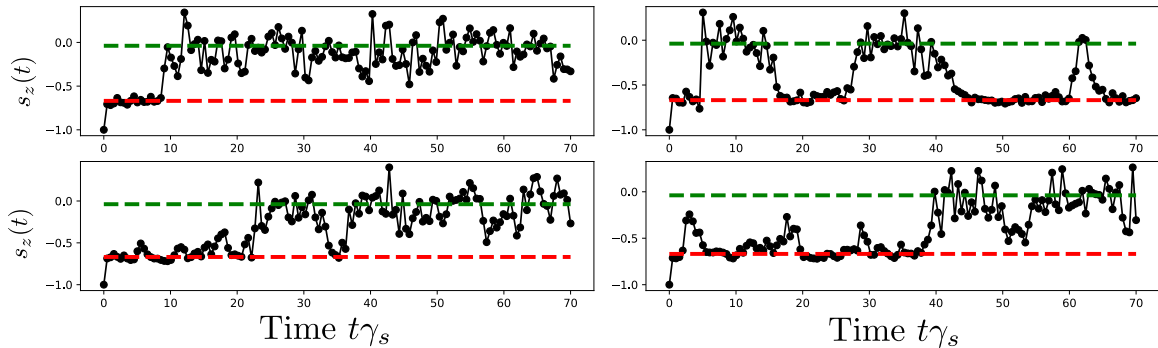
✓ Saves memory: $2^N \times 2^N$ elements

✗ Loses Time: Need to average over many trajectories to get the density matrix.

In our work we found that a bistability is seen in a single trajectory, but the bistability is lost in the density matrix average.

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How typical trajectory looks like (in bistability case)

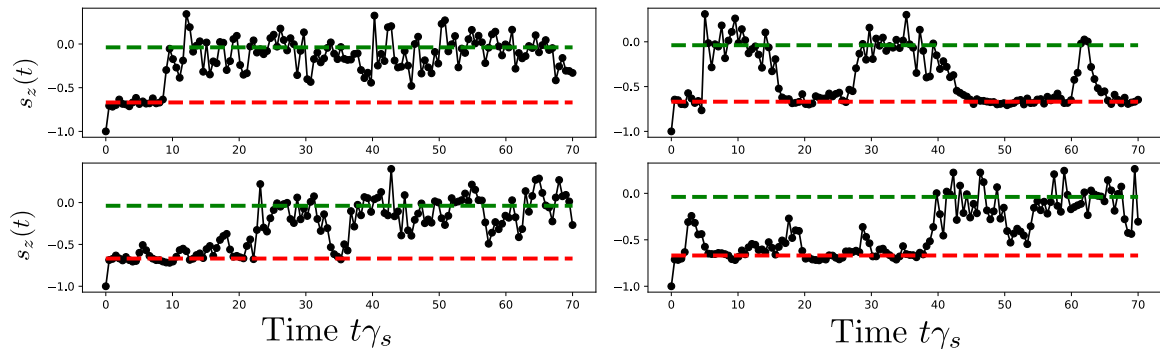


4 examples of quantum trajectories $N = 18$ (best I can get), $\Omega = 0.73\Omega_c$ (bistability region), $\gamma_c = 10\gamma_s$. Red and Green lines are the mean-field solutions. Generated with QuTiP (python library for quantum simulations).

Note: average of $N_{tr} \approx 100$ trajectories gives the density matrix solution, which is a smooth curve (no bistability).

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Problem statement

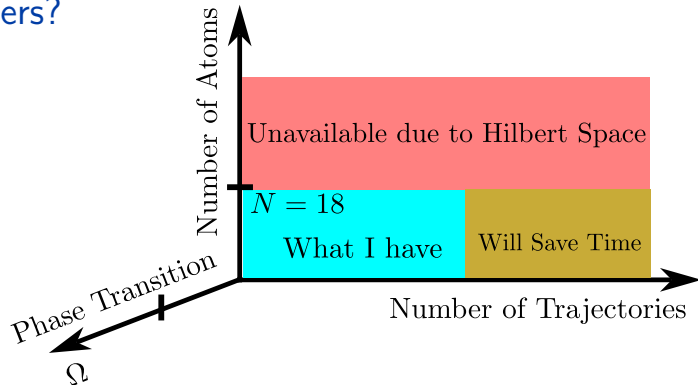


4 examples of quantum trajectories $N = 18$ (best I can get), $\Omega = 0.73\Omega_c$ (bistability region), $\gamma_c = 10\gamma_s$.

Goal of the project

Create a model to generate trajectories similar to quantum trajectories generated by Monte Carlo wave-function simulations in QuTiP

Why this matters?

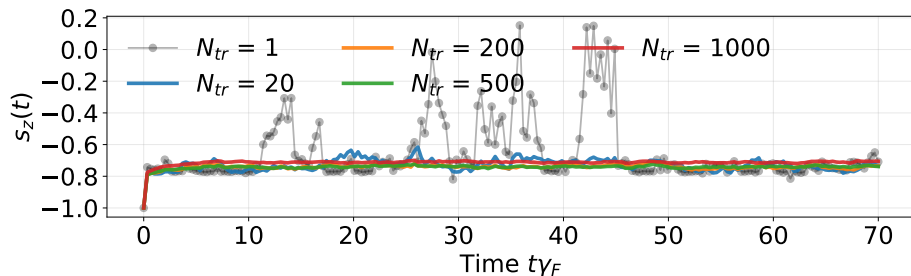


Directions

- Fix N and Ω , generate more trajectories $\Rightarrow$ capture mean and/or bistability
- Fix N , generate trajectories for different $\Omega \Rightarrow$ extrapolate to unseen Ω values
- Full model with N and Ω

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Dataset & Model for $N = 8$

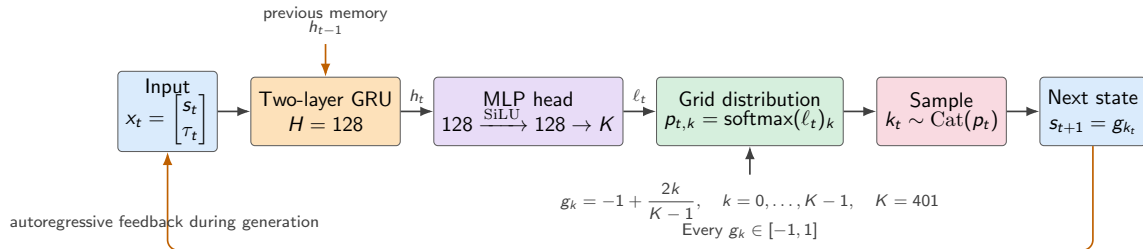


Generated trajectory and averages for $N = 8$ and $\Omega = 0.73\Omega_c$. Dataset contains $N_{tr} = 1000$ trajectories, each with $T = 210$ time steps. We split it into training, testing and validation sets with ratio 80:10:10.

We use a Gate Recurrent Unit (GRU) model for predicting time series on **discrete** grid

$x_{t+1} = f_\theta(x_t|h_t)$; h_t - hidden state. Input: $x_t = [s_t, \tau_t]$; Output: $s_{t+1} \in [-1, 1]$. ✓

TrajectoryGRUGrid: Model Architecture



Training at every time step

$$y_t = s_{t+1}^{\text{real}}, \quad q_{t,i} = 1 - \alpha, \quad q_{t,i+1} = \alpha, \quad \mathcal{L}_t = - \sum_{k=0}^{K-1} q_{t,k} \log p_{t,k}.$$

$\underbrace{\text{real } s_t \text{ as input}}_{\text{teacher forcing}} \longrightarrow \underbrace{\text{two-bin interpolated target } q_t}_{\text{continuous data on a discrete grid}} \longrightarrow \underbrace{\text{categorical cross-entropy}}_{\text{maximize next-state likelihood}}.$

Dataset: Quantum Trajectories with Variable N and Ω

Data Generation (QuTiP MCWF)

Monte Carlo wave-function simulations of a spin chain with dissipation:

- Chain of N qubits with XY drive: $\mathcal{H} = 2\Omega\hat{S}_x$
- Dephasing: $\gamma_0 = 1.0$ per qubit, collective dissipation: $\gamma_c = 10.0$
- Time range: $t \in [0, 70]$ with 210 time points
- Observable: collective spin $S_z = \sum_n \sigma_z^{(n)}$, normalized by N

Dataset Structure

Multiple datasets with varying conditions:

- **Particle numbers:** $N \in \{4, 6, 8, 10, 12, 14, 16, 18\}$
- **Drive ratios:** $\Omega/\Omega_c \in \{0.61, 0.63, 0.65, 0.67, 0.69, 0.71, 0.75\}$ (bistability region)
- **Trajectories per condition:** ~ 1000 stochastic realizations
- **Split:** Train (80%), Validation (10%), Test (10%) *stratified by* (N, Ω)